

HW_Group1

Prediction model for mushroom dataset

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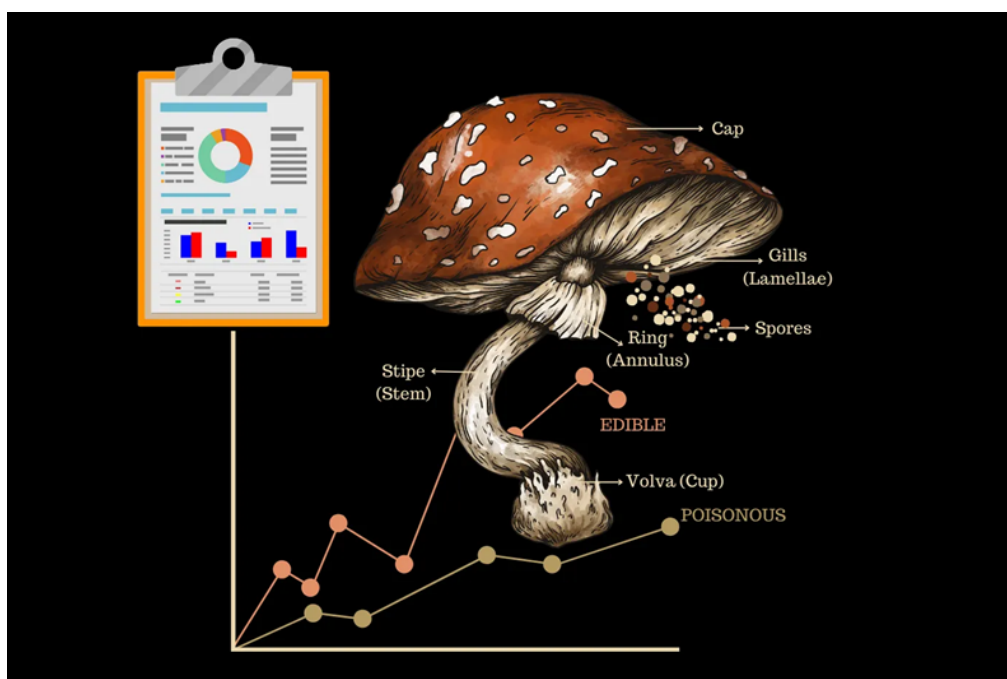
一、Data Dictionary

表 1: Mushroom Dataset Data Dictionary

Variable	DataType	Definition	Note
family	String	Name of the family of mushroom species	Multinomial
name	String	Mushroom species name	Multinomial
class	Binary	poisonous=p, edible=e	Binary

Variable	Data Type	Definition	Note
cap-diameter	Float, Metric data	Cap diameter in cm	[Min, max] or mean
cap-shape	Nominal data	Shape of the cap	bell=b, conical=c, convex=x, flat=f, sunken=s, spherical=p, others=o
cap-surface	Nominal data	Surface type of the cap	fibrous=i, grooves=g, scaly=y, smooth=s, shiny=h, leathery=l, silky=k, sticky=t, wrinkled=w, fleshy=e
cap-color	Nominal data	Color of the cap	brown=n, buff=b, gray=g, green=r, pink=p, purple=u, red=e, white=w, yellow=y, blue=l, orange=o, black=k
does-bruise-bleed	Nominal data	Bruising or bleeding	t=yes, f=no
gill-attachment	Nominal data	Attachment of the gills	adnate=a, adnexed=x, decurrent=d, free=e, sinuate=s, pores=p, none=f, unknown=?
gill-spacing	Nominal data	Spacing between gills	close=c, distant=d, none=f
gill-color	Nominal data	Color of the gills	brown=n, buff=b, gray=g, green=r, pink=p, purple=u, red=e, white=w, yellow=y, blue=l, orange=o, black=k, none=f
stem-height	Float, Metric data	Height of the stem in cm	[Min, max] or mean
stem-width	Float, Metric data	Width of the stem in mm	[Min, max] or mean
stem-root	Nominal data	Root type of the stem	bulbous=b, swollen=s, club=c, cup=u, equal=e, rhizomorphs=z, rooted=r
stem-surface	Nominal data	Surface type of the stem	fibrous=i, grooves=g, scaly=y, smooth=s, shiny=h, leathery=l, silky=k, sticky=t, wrinkled=w, fleshy=e, none=f
stem-color	Nominal data	Color of the stem	brown=n, buff=b, gray=g, green=r, pink=p, purple=u, red=e, white=w, yellow=y, blue=l, orange=o, black=k, none=f
veil-type	Nominal data	Type of veil	p=partial, u=universal
veil-color	Nominal data	Color of the veil	brown=n, buff=b, gray=g, green=r, pink=p, purple=u, red=e, white=w, yellow=y, blue=l, orange=o, black=k, none=f

Variable	DataType	Definition	Note
has-ring	Nominal data	Presence of a ring	t=yes, f=no
ring-type	Nominal data	Type of ring	cobwebby=c, evanescent=e, flaring=r, grooved=g, large=l, pendant=p, sheathing=s, zone=z, scaly=y, movable=m, none=f, unknown=?
spore-print-color	Nominal data	Color of spore-print	brown=n, buff=b, gray=g, green=r, pink=p, purple=u, red=e, white=w, yellow=y, blue=l, orange=o, black=k
habitat	Nominal data	Habitat type	grasses=g, leaves=l, meadows=m, paths=p, heaths=h, urban=u, waste=w, woods=d
season	Nominal data	Season of occurrence	spring=s, summer=u, autumn=a, winter=w



二、讀取資料

```
# R Interface to Python
library(reticulate) # Make R and Python interoperable

# Finding Anaconda's Python path
use_python("C:/Users/user/anaconda3/python.exe", required = TRUE)
```

```
# py_config()

library(Hmisc)           # data analysis and report tools
library(ggplot2)         # a system for creating graphics
library(tableone)        # a tool for creating tableone

# install.packages("caret")
# install.packages("ggplot2")
# install.packages("brglm2")
# install.packages("randomForest")
# install.packages("xgboost")

# !pip install tableone
# ! pip install scikit-learn
# ! pip install tensorflow
# ! pip install lightgbm
# ! pip install imblearn

import re # Regular expressions (text processing)
import numpy as np # Numerical computing
import pandas as pd # Data analysis
import seaborn as sns # Data visualization
import tensorflow as tf # Deep learning framework
from tensorflow import keras # High-level API for TensorFlow
from tableone import TableOne # Summary tables for data analysis
import matplotlib.pyplot as plt # Plotting library
from tensorflow.keras import layers, regularizers # Keras layers & regularization
# Split dataset and calculate val score
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.preprocessing import LabelEncoder # Encode categorical data
from sklearn.preprocessing import StandardScaler # Standardize features
from sklearn.metrics import accuracy_score, classification_report # Model evaluation
from sklearn.ensemble import RandomForestClassifier # Machine learning classifier
from tensorflow.keras import backend as K # Low-level TensorFlow operations
from xgboost import XGBClassifier # XGBoost 分類器套件
from lightgbm import LGBMClassifier # LightGBM 分類器套件
from sklearn.linear_model import LogisticRegression # 邏輯回歸套件
```

三、資料前處理-轉換為結構化資料

```
# read CSV
file_path = "C:/Users/user/Downloads/primary_data.csv"

with open(file_path, "r", encoding="utf-8") as file:
    raw_lines = file.readlines()

# get the feature name
header = raw_lines[0].strip().split(";") # the first row

# processing data
data = []
for line in raw_lines[1:]:
    values = line.strip().split(";") # spilt by `;`
    if len(values) != len(header):
        values += [""] * (len(header) - len(values))

    # handing `[ ]` to let [10 20] to '10, 20'
    cleaned_values = [re.sub(r"\[|\]", "", v).replace("\t", " ") for v in values]
    data.append(cleaned_values)

# make a DataFrame
df1 = pd.DataFrame(data, columns=header)
```

四、資料描述

```
# read dataset
df <- read.csv("C:/Users/user/Downloads/primary_data_cleaned.csv")
# data description
latex(describe(df), descript = "descriptive statistics", file = '', caption.placement = 'top')
```

23 Variables			df 173 Observations	
family				
n	missing	distinct		
173	0	23		
lowest :	Amanita Family	Bolbitius Family	Bolete Family	Bracket Fungi
highest:	Russula Family	Saddle-Cup Family	Stropharia Family	Tricholoma Family
				Chanterelle Family
				Wax Gill Family

name

n	missing	distinct
173	0	173

lowest : Amethyst Deceiver	Aniseed Funnel Cap	Apricot Fungus	Bare-toothed Russula	Bay Bolete
highest: Yellow-gilled Russula	Yellow-staining Mushroom	Yellow-stemmed Bell Cap	Yellow Swamp Russula	Yellow Wax cap

class

n	missing	distinct
173	0	2

Value	e	p
Frequency	77	96
Proportion	0.445	0.555

cap.diameter

n	missing	distinct
173	0	51

lowest : 0.4, 1 0.5, 1 0.5, 1.5 0.7, 1.3 1, 1.5 , highest: 8, 14 8, 15 8, 20 8, 25 8, 30

cap.shape

n	missing	distinct
173	0	27

lowest : b b, f b, f, s b, x b, x, f, highest: x, f x, f, s x, o x, p x, s

Cap.surface

n	missing	distinct
133	40	40

lowest : d d, e, y, i d, k d, k, s d, s
highest: t, w, d w w, t y y, s

cap.color

n	missing	distinct
173	0	67

lowest : b b, p, e, y b, u e, n
highest: y y, n y, o y, o, g, n, r y, o, r, n

does.bruise.or.bleed

n	missing	distinct
173	0	2

Value	f	t
Frequency	143	30
Proportion	0.827	0.173

gill_attachment

n	missing	distinct
145	28	8

Value	a	a, d	d	e	f	p	s	x
Frequency	32	8	25	16	10	17	16	21
Proportion	0.221	0.055	0.172	0.110	0.069	0.117	0.110	0.145

gill_spacing

n	missing	distinct
102	71	3

Value	c	d	f
Frequency	70	22	10
Proportion	0.686	0.216	0.098

gill.color

n	missing	distinct
173	0	59

lowest : b b, p, w b, u e f , highest: y, n y, o, e y, r y, r, k y, w

stem.height

n	missing	distinct
173	0	46

lowest : 0 1, 2 1, 3 10, 12 10, 15, highest: 8, 12 8, 15 8, 20 8, 25 8, 30

stem.width

n	missing	distinct
173	0	48

lowest : 0 0.5, 1 1 1, 2 1, 3 , highest: 7, 15 8, 12 8, 15 8, 18 8, 20

stem_root

n	missing	distinct
27	146	5

Value	b	c	f	r	s
Frequency	9	2	3	4	9
Proportion	0.333	0.074	0.111	0.148	0.333

stem.surface

n	missing	distinct
65	108	14

Value	f	g	h	i	i, s	i, t	i, y	k	k, s	s	s, h	t	y	y, s
Frequency	3	5	1	11	1	1	1	4	1	15	1	7	13	1
Proportion	0.046	0.077	0.015	0.169	0.015	0.015	0.015	0.062	0.015	0.231	0.015	0.108	0.200	0.015

stem.color

n	missing	distinct
173	0	41

lowest : b, u e e, n e, u, y e, y , highest: w, y y y, e, n y, n y, o, k

veil.type

n	missing	distinct	value
9	164	1	u

Value	u
Frequency	9
Proportion	1

veil.color

n	missing	distinct
21	152	7

Value	e, n	k	n	u	w	y	y, w
Frequency	1	1	1	1	15	1	1
Proportion	0.048	0.048	0.048	0.048	0.714	0.048	0.048

has.ring

n	missing	distinct
173	0	2

Value	f	t
Frequency	130	43
Proportion	0.751	0.249

	e	p	Overall
	(N=77)	(N=96)	(N=173)
gill_attachment			
	10 (13.0%)	18 (18.8%)	28 (16.2%)
a	11 (14.3%)	21 (21.9%)	32 (18.5%)
a, d	5 (6.5%)	3 (3.1%)	8 (4.6%)
d	9 (11.7%)	16 (16.7%)	25 (14.5%)
e	10 (13.0%)	6 (6.3%)	16 (9.2%)
f	4 (5.2%)	6 (6.3%)	10 (5.8%)
p	12 (15.6%)	5 (5.2%)	17 (9.8%)
s	7 (9.1%)	9 (9.4%)	16 (9.2%)
x	9 (11.7%)	12 (12.5%)	21 (12.1%)
gill_spacing			
	31 (40.3%)	40 (41.7%)	71 (41.0%)
c	29 (37.7%)	41 (42.7%)	70 (40.5%)
d	13 (16.9%)	9 (9.4%)	22 (12.7%)
f	4 (5.2%)	6 (6.3%)	10 (5.8%)
stem_root			
	67 (87.0%)	79 (82.3%)	146 (84.4%)
b	6 (7.8%)	3 (3.1%)	9 (5.2%)
s	4 (5.2%)	5 (5.2%)	9 (5.2%)
c	0 (0%)	2 (2.1%)	2 (1.2%)
f	0 (0%)	3 (3.1%)	3 (1.7%)
r	0 (0%)	4 (4.2%)	4 (2.3%)
habitat			
d	47 (61.0%)	57 (59.4%)	104 (60.1%)
d, h	1 (1.3%)	3 (3.1%)	4 (2.3%)
g	1 (1.3%)	10 (10.4%)	11 (6.4%)
g, d	6 (7.8%)	4 (4.2%)	10 (5.8%)
g, d, h	1 (1.3%)	0 (0%)	1 (0.6%)
g, h, d	1 (1.3%)	2 (2.1%)	3 (1.7%)
g, l, m, d	1 (1.3%)	0 (0%)	1 (0.6%)
g, m	3 (3.9%)	2 (2.1%)	5 (2.9%)
g, m, d	1 (1.3%)	4 (4.2%)	5 (2.9%)
g, u, d	1 (1.3%)	0 (0%)	1 (0.6%)
l	1 (1.3%)	0 (0%)	1 (0.6%)
l, d	7 (9.1%)	6 (6.3%)	13 (7.5%)
l, d, h	1 (1.3%)	0 (0%)	1 (0.6%)
l, h	1 (1.3%)	0 (0%)	1 (0.6%)
m	1 (1.3%)	1 (1.0%)	2 (1.2%)
m, d	2 (2.6%)	1 (1.0%)	3 (1.7%)
w	1 (1.3%)	0 (0%)	1 (0.6%)

六、視覺化圖表分析

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# read CSV
df = pd.read_csv("C:/Users/user/Downloads/primary_data_cleaned.csv")

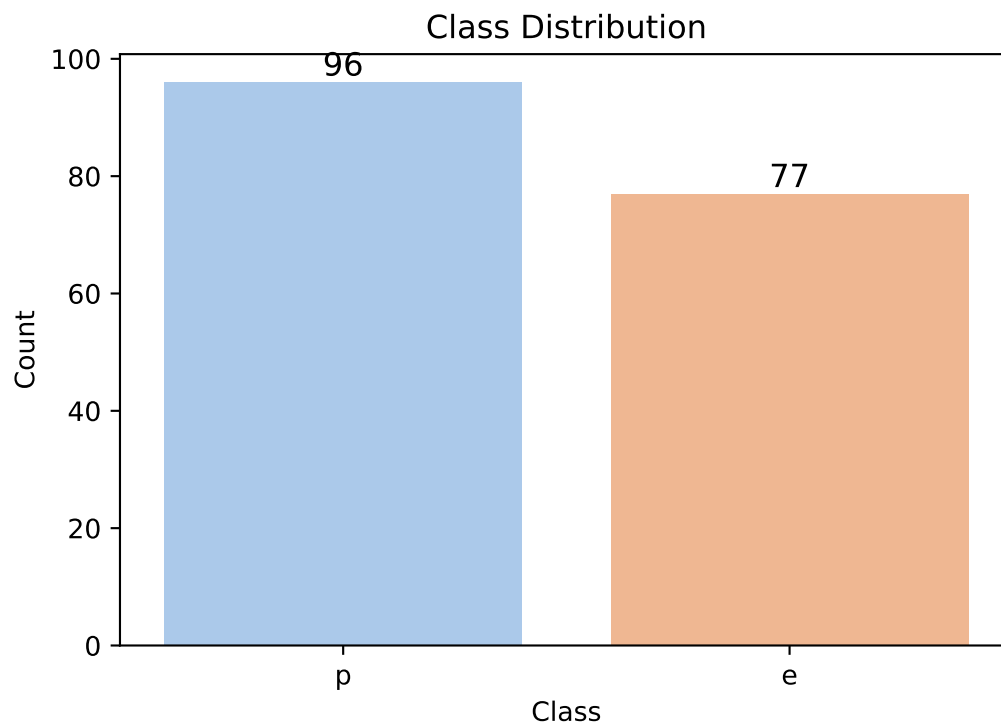
# print class's counts
print(df['class'].value_counts())

class
p    96
e    77
Name: count, dtype: int64

# make class distribution
plt.figure(figsize=(6, 4))
ax = sns.countplot(data=df, x="class", palette="pastel")
plt.title("Class Distribution")
plt.xlabel("Class")
plt.ylabel("Count")

# add the number on the chart
for p in ax.patches:
    plt.text(p.get_x() + p.get_width() / 2., p.get_height(),
             f'{p.get_height():.0f}',
             ha='center', va='bottom', fontsize=12, color='black')

# show the chart
plt.show()
```



from this chart,we can see that there are 96 poisonous and 77 edible.

```
library(GGally)
library(ggplot2)

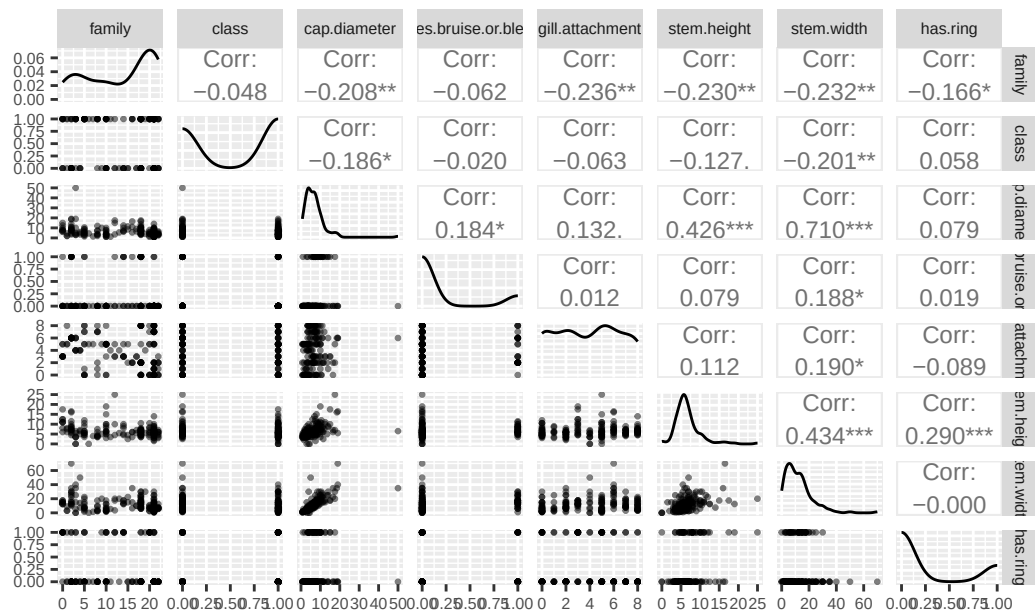
# read CSV
df <- read.csv("C:/Users/user/Downloads/processed_data11.csv")

# select the numerical_columns
numerical_columns <- sapply(df, is.numeric)

# create a ggpairs chart
plot <- ggpairs(df[, numerical_columns],
               upper = list(continuous = wrap("cor", size = 3)), # cor-plot size
               lower = list(continuous = wrap("points", alpha = 0.5, size = 0.5)), # pointsize
               title = "GGpairs Plot") +
  theme(text = element_text(size = 8)) # textsize

# show the plot
print(plot)
```

GGpairs Plot



```
import matplotlib.pyplot as plt
import seaborn as sns

# select the categorical_columns
categorical_columns = ['family', 'class', 'cap-diameter', 'cap-shape', 'Cap-surface', 'cap-color',
                      'does-bruise-or-bleed', 'gill_attachment', 'gill_spacing', 'gill-color',
                      'stem-height', 'stem-width', 'stem-surface', 'stem-color', 'has-ring',
                      'ring-type', 'habitat', 'season']

# df1 = pd.read_csv("C:/Users/user/Downloads/primary_data_cleaned.csv")

# create a 6x3 chart
fig, axes = plt.subplots(nrows=6, ncols=3, figsize=(80, 120)) # chartsize
axes = axes.flatten() # Flatten a 2D array to 1D

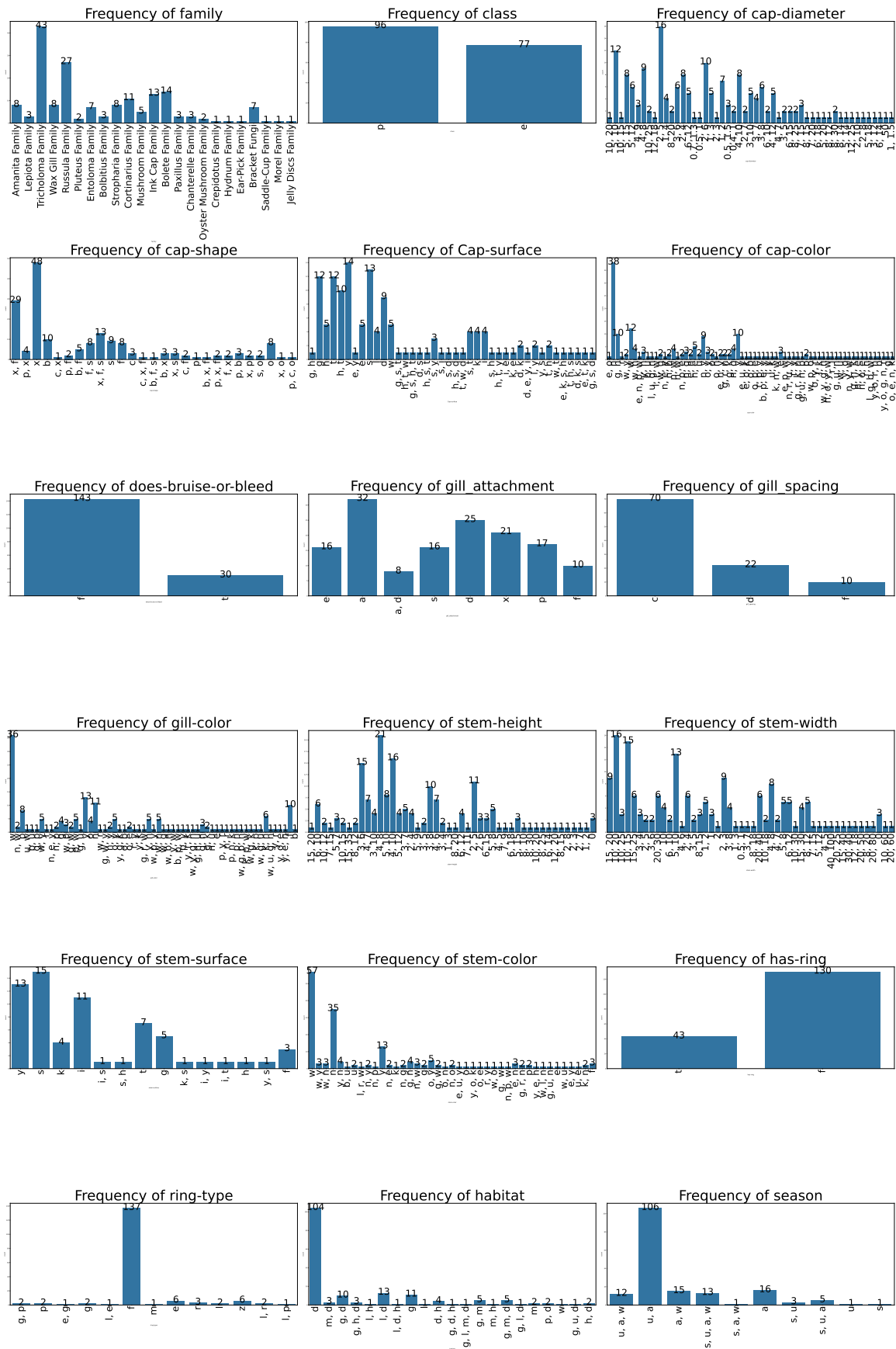
# create all the chart
for i, column in enumerate(categorical_columns):
    ax = sns.countplot(data=df, x=column, ax=axes[i])
    ax.set_title(f'Frequency of {column}', fontsize=90) # set title
    ax.tick_params(axis='x', rotation=90, labels=60) # rotation and labels

# add the number on the chart
for p in ax.patches:
```

```
ax.annotate(f'{int(p.get_height())}',
            (p.get_x() + p.get_width() / 2., p.get_height()),
            ha='center', va='center',
            fontsize=60, color='black',
            xytext=(0, 5), textcoords='offset points')

# Adjust layout
plt.tight_layout()

# show the chart
plt.show()
```



七、資料前處理-補值 +encoding

```
df = pd.read_csv("C:/Users/user/Downloads/primary_data_cleaned.csv")
# calcute the missing values
missing_values = df.isnull().sum()
print(" 缺失值統計")
```

缺失值統計

```
print(missing_values[missing_values > 0])
```

```
Cap-surface          40
gill_attachment      28
gill_spacing         71
stem_root           146
stem-surface         108
veil-type            164
veil-color           152
ring-type             7
Spore-print-color    155
dtype: int64
```

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import LabelEncoder
from sklearn.experimental import enable_iterative_imputer
from sklearn.impute import IterativeImputer

df=pd.read_csv("C:/Users/user/Downloads/primary_data_cleaned.csv")
# mice imputation
# select the numerical feature
numerical_columns = ['cap-diameter', 'stem-height', 'stem-width']

def convert_range_to_median(value):
    if pd.isna(value):
        return np.nan
    if isinstance(value, str) and ',' in value:
        try:
            values = [float(x.strip()) for x in value.split(',')]
            return np.median(values)
```

```
        except:
            return np.nan
    try:
        return float(value)
    except:
        return np.nan

# convert to median
for col in numerical_columns:
    df[col] = df[col].apply(convert_range_to_median)

# use MICE stagety
imputer = IterativeImputer(random_state=0)
df[numerical_columns] = imputer.fit_transform(df[numerical_columns])

# Label Encoding
label_columns = ['family', 'class', 'does-bruise-or-bleed', 'gill_attachment', 'has-ring']
label_encoder = LabelEncoder()

for col in label_columns:
    # if it have nan, change it to missing
    df[col] = df[col].fillna('missing')
    df[col] = label_encoder.fit_transform(df[col].astype(str))

# delete the missing feature
columns_to_drop = ['veil-type', 'veil-color', 'Spore-print-color', 'stem-surface',
                  'stem_root', 'name']
df.drop(columns=[col for col in columns_to_drop if col in df.columns], inplace=True)

# 4. One-Hot Encoding
multi_columns = ['cap-shape', 'Cap-surface', 'cap-color', 'gill_spacing', 'gill-color',
                 'stem-color', 'ring-type', 'season', 'habitat']

for col in multi_columns:
    if col in df.columns:
        # fill in nan
        df[col] = df[col].fillna('').astype(str) # 確保轉成字串
        # split the multi column
        dummies = df[col].str.split(',', expand=True).stack().str.strip()
```



```
dummies = pd.get_dummies(dummies, prefix=col)
# groupby it
dummies = dummies.groupby(level=0).sum()
# concat to DataFrame
df = pd.concat([df, dummies], axis=1)
# drop the original column
df.drop(columns=[col], inplace=True)

# know that how size is it
print(" 資料大小 :", df.shape)
```

資料大小 : (173, 89)

```
# get the processed_data1
df.to_csv('processed_data1.csv', index=False)
```

I first use MICE (IterativeImputer) to fill in the values. MICE (Multiple Imputation by Chained Equations) can infer missing values based on other numerical features, which is more accurate than simple mean filling.

Then because of cap-diameter, stem-height, stem-width Some of these numerical features are in range format, so I designed a function convert_range_to_median for subsequent processing. If it is a range (, separated), the median is calculated to fill the value. If it cannot be parsed, it is set to NaN to facilitate subsequent value filling.

Next, we use Label Encoding to process categorical data because they have fewer categories, such as family, class, does-bruise-or-bleed, gill-attachment, has-ring. If there are NaN values, they will be filled with 'missing' to avoid LabelEncoder errors.

Then we remove unimportant features or features with too many missing values, such as: veil-type, veil-color, pore-print-color, stem-surface, stem_root, name → These features may not contain enough information or cannot be directly quantified, so they are removed to simplify the model.

One-Hot Encoding is also used. The processing method for this part is special because some features (such as cap-shape, cap-surface, cap-color, gill-spacing...) may be multi-valued (comma-separated), for example: cap-color, season, etc. Therefore, I did the following steps to fill NaN with "" (empty string) to avoid str.split() errors. str.split(' ') splits into multiple categories and then expands into multiple lines (stack()). Use pd.get_dummies() for One-Hot Encoding and convert it to 0/1 features. Use groupby(level=0).sum() to merge multiple rows of results back into the original DataFrame. Delete the original feature field to avoid duplication.

Finally, output the data size (df.shape) to check the final number of features and confirm

whether One-Hot Encoding successfully added new features. Save it as processed_data1.csv for subsequent modeling.

八、模型訓練比較

```
# read dataset
df <- read.csv("C:/Users/user/Downloads/processed_data1.csv")

model <- lm(class ~., df)
summary(model)
```

Call:

```
lm(formula = class ~ ., data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.97340	-0.25092	0.00303	0.26157	0.90276

Coefficients: (2 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.3615244	0.9065119	1.502	0.1368
family	-0.0038817	0.0097724	-0.397	0.6922
cap.diameter	-0.0189028	0.0165759	-1.140	0.2573
does.bruise.or.bleed	-0.0994657	0.1516359	-0.656	0.5136
gill_attachment	-0.0209536	0.0216581	-0.967	0.3360
stem.height	0.0303277	0.0291469	1.041	0.3010
stem.width	-0.0050993	0.0090648	-0.563	0.5752
has.ring	-0.1944750	0.2993648	-0.650	0.5177
cap.shape_b	0.0623533	0.2091215	0.298	0.7663
cap.shape_c	-0.4271527	0.2738616	-1.560	0.1225
cap.shape_f	0.0482698	0.1271345	0.380	0.7051
cap.shape_o	-0.1250666	0.3205233	-0.390	0.6974
cap.shape_p	-0.1106911	0.2275592	-0.486	0.6279
cap.shape_s	-0.0266050	0.1688331	-0.158	0.8752
cap.shape_x	-0.1722741	0.1460528	-1.180	0.2414
Cap.surface_	-0.0524850	0.1749313	-0.300	0.7649
Cap.surface_d	-0.1077828	0.1950522	-0.553	0.5820
Cap.surface_e	0.0578545	0.2233436	0.259	0.7962

Cap.surface_g	0.0020091	0.2137696	0.009	0.9925
Cap.surface_h	-0.2149626	0.1773082	-1.212	0.2287
Cap.surface_i	0.0978122	0.2611298	0.375	0.7089
Cap.surface_k	0.4681604	0.2482930	1.886	0.0627 .
Cap.surface_l	-0.2045603	0.3721132	-0.550	0.5839
Cap.surface_s	-0.1277217	0.1543667	-0.827	0.4103
Cap.surface_t	0.1669481	0.1726111	0.967	0.3362
Cap.surface_w	-0.2280771	0.2622338	-0.870	0.3869
Cap.surface_y	-0.0058526	0.2101227	-0.028	0.9778
cap.color_b	-0.0567223	0.2877796	-0.197	0.8442
cap.color_e	0.3522773	0.1823324	1.932	0.0566 .
cap.color_g	0.2870923	0.1832476	1.567	0.1209
cap.color_k	0.0755929	0.2540775	0.298	0.7668
cap.color_l	-0.3499466	0.3622141	-0.966	0.3367
cap.color_n	-0.2891329	0.1344178	-2.151	0.0343 *
cap.color_o	0.0945935	0.1987237	0.476	0.6353
cap.color_p	-0.1989331	0.2379394	-0.836	0.4054
cap.color_r	0.3700774	0.2488053	1.487	0.1406
cap.color_u	-0.3346817	0.2898931	-1.155	0.2515
cap.color_w	-0.0549760	0.1495666	-0.368	0.7141
cap.color_y	-0.0098968	0.1448291	-0.068	0.9457
gill.spacing_	-0.0396271	0.3395515	-0.117	0.9074
gill.spacing_c	-0.0371681	0.3602963	-0.103	0.9181
gill.spacing_d	-0.1044233	0.3787073	-0.276	0.7834
gill.spacing_f	NA	NA	NA	NA
gill.color_b	-0.0446901	0.3703743	-0.121	0.9042
gill.color_e	-0.0002124	0.3094818	-0.001	0.9995
gill.color_f	NA	NA	NA	NA
gill.color_g	-0.1611256	0.1682301	-0.958	0.3409
gill.color_k	0.0479289	0.2257801	0.212	0.8324
gill.color_n	0.1379421	0.1419308	0.972	0.3338
gill.color_o	0.1604546	0.2539942	0.632	0.5292
gill.color_p	0.0512740	0.1670390	0.307	0.7596
gill.color_r	0.1965992	0.2865332	0.686	0.4945
gill.color_u	0.3606675	0.3297526	1.094	0.2771
gill.color_w	0.0836331	0.1586876	0.527	0.5995
gill.color_y	0.0333993	0.1655329	0.202	0.8406
stem.color_b	-0.3430223	0.8705746	-0.394	0.6945
stem.color_e	-0.2357548	0.2460527	-0.958	0.3407
stem.color_f	0.4258553	0.5212757	0.817	0.4162

stem.color_g	-0.3159341	0.2321426	-1.361	0.1771
stem.color_k	0.1806556	0.3884494	0.465	0.6431
stem.color_l	-0.1408029	0.6781277	-0.208	0.8360
stem.color_n	0.0456725	0.1632934	0.280	0.7804
stem.color_o	-0.2593299	0.2527932	-1.026	0.3078
stem.color_p	0.4949640	0.3437107	1.440	0.1535
stem.color_r	-0.0838529	0.4535706	-0.185	0.8538
stem.color_u	0.3250844	0.3485060	0.933	0.3535
stem.color_w	-0.2952267	0.1824844	-1.618	0.1094
stem.color_y	-0.0896282	0.1986856	-0.451	0.6530
ring.type_	-0.1310358	0.4946129	-0.265	0.7917
ring.type_e	0.0203098	0.4507510	0.045	0.9642
ring.type_f	-0.1324176	0.5802433	-0.228	0.8200
ring.type_g	-0.0166983	0.4745727	-0.035	0.9720
ring.type_l	0.0746727	0.3851254	0.194	0.8467
ring.type_m	-0.8731507	0.9850602	-0.886	0.3779
ring.type_p	0.1843888	0.4583599	0.402	0.6885
ring.type_r	0.2314084	0.4745367	0.488	0.6270
ring.type_z	0.4252290	0.5150430	0.826	0.4113
season_a	-0.0311993	0.3614524	-0.086	0.9314
season_s	-0.0868219	0.1953683	-0.444	0.6579
season_u	0.0092059	0.1527630	0.060	0.9521
season_w	-0.2388900	0.1672120	-1.429	0.1567
habitat_d	-0.0968583	0.1910066	-0.507	0.6134
habitat_g	0.0176560	0.1476305	0.120	0.9051
habitat_h	-0.0690233	0.1880134	-0.367	0.7144
habitat_l	-0.1137950	0.1883968	-0.604	0.5474
habitat_m	-0.0264021	0.2241409	-0.118	0.9065
habitat_p	0.6531000	0.5446421	1.199	0.2338
habitat_u	-0.5817007	0.7773679	-0.748	0.4563
habitat_w	-0.8194416	0.7517710	-1.090	0.2788

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5076 on 86 degrees of freedom

Multiple R-squared: 0.4815, Adjusted R-squared: -0.03708

F-statistic: 0.9285 on 86 and 86 DF, p-value: 0.6342

在原始的未進行預處理的資料中包含 89 個變數，經由 linear models 訓練，從結果可見，Multiple R-squared 僅有 0.4815，預測表現不是那麼好。

```
library(MASS)
library(caret)
library(detectseparation)

# 讀取資料
df <- read.csv("C:/Users/user/Downloads/processed_data1.csv")
df$class <- as.factor(df$class)

# 分割資料
set.seed(66)
trainIndex <- createDataPartition(df$class, p = 0.7, list = FALSE)
train_data <- df[trainIndex, ]
test_data <- df[-trainIndex, ]

# 合併稀疏層次 ( 出現次數少於 5 次的層次合併為 "Other" )
for (col in names(train_data)) {
  if (is.factor(train_data[[col]]) && col != "class") {
    freq <- table(train_data[[col]])
    train_data[[col]] <- as.character(train_data[[col]])
    train_data[[col]][freq[train_data[[col]]] < 5] <- "Other"
    train_data[[col]] <- as.factor(train_data[[col]])

    test_data[[col]] <- as.character(test_data[[col]])
    test_data[[col]][freq[test_data[[col]]] < 5] <- "Other"
    test_data[[col]] <- as.factor(test_data[[col]])
  }
}

# 檢查分離問題
# 提取響應變數 y 和設計矩陣 x
y <- as.numeric(train_data$class) - 1 # 將 class 轉為 0/1 ( 二項分佈需要 )
x <- model.matrix(class ~ . - 1, data = train_data) # 不包含截距

# 使用 detect_separation
separation_result <- detect_separation(x = x, y = y, family = binomial())
print(separation_result)
```

Implementation: ROI | Solver: lpSolve

Separation: TRUE

Existence of maximum likelihood estimates

family	cap.diameter	does.bruise.or.bleed
Inf	-Inf	-Inf
gill_attachment	stem.height	stem.width
-Inf	Inf	-Inf
has.ring	cap.shape_b	cap.shape_c
-Inf	Inf	-Inf
cap.shape_f	cap.shape_o	cap.shape_p
-Inf	Inf	-Inf
cap.shape_s	cap.shape_x	Cap.surface_
Inf	-Inf	Inf
Cap.surface_d	Cap.surface_e	Cap.surface_g
-Inf	Inf	-Inf
Cap.surface_h	Cap.surface_i	Cap.surface_k
-Inf	Inf	Inf
Cap.surface_l	Cap.surface_s	Cap.surface_t
Inf	Inf	Inf
Cap.surface_w	Cap.surface_y	cap.color_b
Inf	-Inf	Inf
cap.color_e	cap.color_g	cap.color_k
Inf	Inf	Inf
cap.color_l	cap.color_n	cap.color_o
-Inf	-Inf	Inf
cap.color_p	cap.color_r	cap.color_u
Inf	Inf	-Inf
cap.color_w	cap.color_y	gill_spacing_
-Inf	Inf	-Inf
gill_spacing_c	gill_spacing_d	gill_spacing_f
-Inf	-Inf	Inf
gill.color_b	gill.color_e	gill.color_f
-Inf	-Inf	NA
gill.color_g	gill.color_k	gill.color_n
Inf	Inf	Inf
gill.color_o	gill.color_p	gill.color_r
Inf	Inf	Inf
gill.color_u	gill.color_w	gill.color_y
Inf	-Inf	Inf
stem.color_b	stem.color_e	stem.color_f
NA	Inf	Inf
stem.color_g	stem.color_k	stem.color_l
-Inf	Inf	-Inf

stem.color_n	stem.color_o	stem.color_p
Inf	Inf	Inf
stem.color_r	stem.color_u	stem.color_w
-Inf	Inf	-Inf
stem.color_y	ring.type_	ring.type_e
Inf	-Inf	Inf
ring.type_f	ring.type_g	ring.type_l
Inf	Inf	-Inf
ring.type_m	ring.type_p	ring.type_r
NA	Inf	-Inf
ring.type_z	season_a	season_s
Inf	Inf	-Inf
season_u	season_w	habitat_d
Inf	-Inf	-Inf
habitat_g	habitat_h	habitat_l
Inf	-Inf	Inf
habitat_m	habitat_p	habitat_u
Inf	Inf	-Inf
habitat_w		
-Inf		

0: finite value, Inf: infinity, -Inf: -infinity

```
# 建模
initial_model <- glm(class ~ ., data = train_data, family = binomial)

# 使用 backward selection
stepwise_bic <- stepAIC(initial_model, direction = "backward", k = log(nrow(train_data)),
  trace = FALSE)
summary(stepwise_bic)
```

Call:

```
glm(formula = class ~ cap.diameter + gill_attachment + stem.height +
  cap.shape_x + Cap.surface_g + Cap.surface_h + Cap.surface_t +
  cap.color_e + cap.color_n + cap.color_r + cap.color_u + gill.color_b +
  gill.color_e + gill.color_n + gill.color_o + gill.color_w +
  stem.color_f + stem.color_l + stem.color_n + stem.color_o +
  stem.color_u + ring.type_e + ring.type_f + ring.type_g +
  ring.type_r + ring.type_z + season_w + habitat_h + habitat_w,
  family = binomial, data = train_data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-462.1	2791009.0	0.000	1.000
cap.diameter	-148.0	8624.2	-0.017	0.986
gill_attachment	-182.7	10316.4	-0.018	0.986
stem.height	104.1	6641.4	0.016	0.987
cap.shape_x	-766.4	43977.0	-0.017	0.986
Cap.surface_g	-868.5	50079.0	-0.017	0.986
Cap.surface_h	-865.5	50139.4	-0.017	0.986
Cap.surface_t	1025.8	58447.0	0.018	0.986
cap.color_e	1752.8	99483.4	0.018	0.986
cap.color_n	-829.2	47473.4	-0.017	0.986
cap.color_r	1843.2	140259.6	0.013	0.990
cap.color_u	-2555.9	583680.1	-0.004	0.997
gill.color_b	-2866.2	161222.7	-0.018	0.986
gill.color_e	-1993.2	112336.5	-0.018	0.986
gill.color_n	944.7	53945.1	0.018	0.986
gill.color_o	4397.8	248572.2	0.018	0.986
gill.color_w	-1257.7	70055.4	-0.018	0.986
stem.color_f	1279.7	166224.6	0.008	0.994
stem.color_l	-3176.5	201129.3	-0.016	0.987
stem.color_n	1467.0	82294.6	0.018	0.986
stem.color_o	-3433.0	194457.5	-0.018	0.986
stem.color_u	1244.7	810312.9	0.002	0.999
ring.type_e	1254.6	2791661.9	0.000	1.000
ring.type_f	2767.7	2795026.0	0.001	0.999
ring.type_g	3342.8	1998470.4	0.002	0.999
ring.type_r	4068.8	3084690.3	0.001	0.999
ring.type_z	5717.4	2809223.5	0.002	0.998
season_w	-2385.4	134349.9	-0.018	0.986
habitat_h	-2245.9	126401.1	-0.018	0.986
habitat_w	-1940.7	1347026.1	-0.001	0.999

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1.6752e+02 on 121 degrees of freedom
 Residual deviance: 7.6958e-07 on 92 degrees of freedom
 AIC: 60

Number of Fisher Scoring iterations: 25

提取選擇的變數

```
selected_vars <- names(coef(stepwise_bic))[-1]
```

```
cat(" 選擇的變數 :", paste(selected_vars, collapse = ", "), "\n")
```

選擇的變數 : cap.diameter, gill_attachment, stem.height, cap.shape_x, Cap.surface_g, Cap.surface_h

最終模型

```
final_formula <- as.formula(paste("class ~", paste(selected_vars, collapse = " + ")))
```

```
final_model <- glm(final_formula, data = train_data, family = binomial)
```

預測

```
test_data$pred_prob <- predict(final_model, newdata = test_data, type = "response")
```

```
test_data$pred_class <- ifelse(test_data$pred_prob > 0.5, levels(df$class)[2], levels(df$class)[1])
```

評估

```
confusion_matrix <- confusionMatrix(factor(test_data$pred_class, levels = levels(df$class)),  
test_data$class)
```

```
print(confusion_matrix)
```

Confusion Matrix and Statistics

	Reference	
Prediction	0	1
0	13	19
1	10	9

Accuracy : 0.4314

95% CI : (0.2935, 0.5775)

No Information Rate : 0.549

P-Value [Acc > NIR] : 0.9660

Kappa : -0.1095

McNemar's Test P-Value : 0.1374

Sensitivity : 0.5652

Specificity : 0.3214

Pos Pred Value : 0.4062

Neg Pred Value : 0.4737

```

Prevalence : 0.4510
Detection Rate : 0.2549
Detection Prevalence : 0.6275
Balanced Accuracy : 0.4433

```

```
'Positive' Class : 0
```

```

accuracy <- confusion_matrix$overall["Accuracy"]
cat(" 預測準確率：", round(accuracy * 100, 2), "%\n")

```

```
預測準確率： 43.14 %
```

我們先經過預處理來合併稀疏層次，並將資料劃分成訓練集跟測試集，比例為 70% 和 30%，接著採用邏輯回歸模型進行二元分類，並使用 `stepAIC()` 進行變數選擇，選擇標準為 BIC。經變數選擇後，模型保留 85 個變數。從模型擬合結果來看，殘差偏差 Residual deviance (1585.92) 遠大於零模型偏差 Null deviance (167.52)，表明模型可能過擬合。預測表現上，模型的準確率為 62.75%，略高於 No Information Rate (54.9%)，但 p 值 (0.1624 > 0.05) 顯示模型表現無顯著差異。Kappa 統計量為 0.2563，表明模型相較於隨機猜測的改進有限 (通常 Kappa > 0.6 表示模型表現較好)。

```

library(caret)
library(randomForest)

# 讀取資料
df <- read.csv("C:/Users/user/Downloads/processed_data1.csv")
df$class <- as.factor(df$class)

# 檢查 class 變數是否包含 NA 或無效值
if (any(is.na(df$class))) {
  cat("class 變數中包含 NA，移除這些觀測值...\n")
  df <- df[!is.na(df$class), ]
}

# 檢查 class 的層次
cat("class 的層次：", levels(df$class), "\n")

```

```
class 的層次： 0 1
```

```

if (length(levels(df$class)) != 2) {
  stop("class 必須是二元變數 ( 僅有兩個層次 )!")
}

```

```
# 分割資料
```

```

set.seed(66)
trainIndex <- createDataPartition(df$class, p = 0.7, list = FALSE)
train_data <- df[trainIndex, ]
test_data <- df[-trainIndex, ]

# 合併稀疏層次 ( 基於 train_data 的頻率 )
for (col in names(train_data)) {
  if (is.factor(train_data[[col]]) && col != "class") {
    freq <- table(train_data[[col]])
    levels_to_keep <- names(freq[freq >= 10])
    train_data[[col]] <- as.character(train_data[[col]])
    train_data[[col]][!train_data[[col]] %in% levels_to_keep] <- "Other"
    train_data[[col]] <- as.factor(train_data[[col]])

    test_data[[col]] <- as.character(test_data[[col]])
    test_data[[col]][!test_data[[col]] %in% levels_to_keep] <- "Other"
    test_data[[col]] <- as.factor(test_data[[col]])

    # 確保因子層次一致
    test_data[[col]] <- factor(test_data[[col]], levels = levels(train_data[[col]]))
  }
}

# 檢查數值變數
numeric_vars <- sapply(train_data, is.numeric)
numeric_data <- train_data[, numeric_vars]

# 檢查標準差和缺失值
std_devs <- apply(numeric_data, 2, sd, na.rm = TRUE)
na_proportions <- colMeans(is.na(numeric_data))
cat(" 標準差 : \n")

```

標準差：

```
print(std_devs)
```

family	cap.diameter	does.bruise.or.bleed
7.58888530	3.84520369	0.37905108
gill_attachment	stem.height	stem.width
2.79813552	3.13360318	9.73871516
has.ring	cap.shape_b	cap.shape_c

0.44604572	0.34773348	0.21713609
cap.shape_f	cap.shape_o	cap.shape_p
0.49526959	0.26247496	0.27544596
cap.shape_s	cap.shape_x	Cap.surface_
0.39915912	0.48448353	0.42225066
Cap.surface_d	Cap.surface_e	Cap.surface_g
0.32003929	0.26247496	0.26247496
Cap.surface_h	Cap.surface_i	Cap.surface_k
0.33895366	0.24855632	0.24855632
Cap.surface_l	Cap.surface_s	Cap.surface_t
0.15551114	0.37905108	0.40530622
Cap.surface_w	Cap.surface_y	cap.color_b
0.19906978	0.34773348	0.23352068
cap.color_e	cap.color_g	cap.color_k
0.33895366	0.37905108	0.23352068
cap.color_l	cap.color_n	cap.color_o
0.19906978	0.48664595	0.29902973
cap.color_p	cap.color_r	cap.color_u
0.23352068	0.27544596	0.24855632
cap.color_w	cap.color_y	gill_spacing_
0.41119677	0.42743283	0.48866019
gill_spacing_c	gill_spacing_d	gill_spacing_f
0.49656727	0.34773348	0.21713609
gill.color_b	gill.color_e	gill.color_f
0.17881272	0.17881272	0.21713609
gill.color_g	gill.color_k	gill.color_n
0.33895366	0.28759758	0.45417272
gill.color_o	gill.color_p	gill.color_r
0.24855632	0.37905108	0.23352068
gill.color_u	gill.color_w	gill.color_y
0.19906978	0.49772549	0.43239627
stem.color_b	stem.color_e	stem.color_f
0.00000000	0.21713609	0.15551114
stem.color_g	stem.color_k	stem.color_l
0.29902973	0.12750670	0.12750670
stem.color_n	stem.color_o	stem.color_p
0.48866019	0.23352068	0.15551114
stem.color_r	stem.color_u	stem.color_w
0.15551114	0.19906978	0.49874521
stem.color_y	ring.type_	ring.type_e

0.39274342	0.19906978	0.23352068
ring.type_f	ring.type_g	ring.type_l
0.41684163	0.17881272	0.17881272
ring.type_m	ring.type_p	ring.type_r
0.00000000	0.15551114	0.17881272
ring.type_z	season_a	season_s
0.19906978	0.15551114	0.35610685
season_u	season_w	habitat_d
0.39274342	0.42743283	0.32003929
habitat_g	habitat_h	habitat_l
0.41119677	0.26247496	0.29902973
habitat_m	habitat_p	habitat_u
0.27544596	0.12750670	0.09053575
habitat_w		
0.09053575		

```
cat("\n缺失值比例：\n")
```

缺失值比例：

```
print(na_proportions)
```

family	cap.diameter	does.bruise.or.bleed
0	0	0
gill_attachment	stem.height	stem.width
0	0	0
has.ring	cap.shape_b	cap.shape_c
0	0	0
cap.shape_f	cap.shape_o	cap.shape_p
0	0	0
cap.shape_s	cap.shape_x	Cap.surface_
0	0	0
Cap.surface_d	Cap.surface_e	Cap.surface_g
0	0	0
Cap.surface_h	Cap.surface_i	Cap.surface_k
0	0	0
Cap.surface_l	Cap.surface_s	Cap.surface_t
0	0	0
Cap.surface_w	Cap.surface_y	cap.color_b
0	0	0

cap.color_e	cap.color_g	cap.color_k
0	0	0
cap.color_l	cap.color_n	cap.color_o
0	0	0
cap.color_p	cap.color_r	cap.color_u
0	0	0
cap.color_w	cap.color_y	gill_spacing_
0	0	0
gill_spacing_c	gill_spacing_d	gill_spacing_f
0	0	0
gill.color_b	gill.color_e	gill.color_f
0	0	0
gill.color_g	gill.color_k	gill.color_n
0	0	0
gill.color_o	gill.color_p	gill.color_r
0	0	0
gill.color_u	gill.color_w	gill.color_y
0	0	0
stem.color_b	stem.color_e	stem.color_f
0	0	0
stem.color_g	stem.color_k	stem.color_l
0	0	0
stem.color_n	stem.color_o	stem.color_p
0	0	0
stem.color_r	stem.color_u	stem.color_w
0	0	0
stem.color_y	ring.type_	ring.type_e
0	0	0
ring.type_f	ring.type_g	ring.type_l
0	0	0
ring.type_m	ring.type_p	ring.type_r
0	0	0
ring.type_z	season_a	season_s
0	0	0
season_u	season_w	habitat_d
0	0	0
habitat_g	habitat_h	habitat_l
0	0	0
habitat_m	habitat_p	habitat_u
0	0	0

```

habitat_w
0

# 移除標準差為零或缺失值比例過高的變數 ( 例如缺失值比例 > 50% )
vars_to_remove <- names(numeric_data)[std_devs == 0 | na_proportions > 0.5]
if (length(vars_to_remove) > 0) {
  cat(" 移除的變數 ( 標準差為零或缺失值過多 ):", vars_to_remove, "\n")
  train_data <- train_data[, !names(train_data) %in% vars_to_remove]
  test_data <- test_data[, !names(test_data) %in% vars_to_remove]
}

```

移除的變數 (標準差為零或缺失值過多) : stem.color_b ring.type_m

```

# 訓練隨機森林模型 ( 直接使用類別變數，無需 one-hot encoding )
rf_model <- randomForest(class ~ ., data = train_data, importance = TRUE)

# 顯示特徵重要性
importance <- importance(rf_model)
cat(" 特徵重要性 : \n")

```

特徵重要性 :

```
print(importance)
```

	0	1	MeanDecreaseAccuracy
family	4.35721184	2.5554103	4.43917130
cap.diameter	3.89860898	1.6443707	3.56665555
does.bruise.or.bleed	-2.56532537	3.0682214	1.00014518
gill_attachment	0.77949218	0.4956533	0.97518113
stem.height	0.35616435	4.0365405	3.05873756
stem.width	4.86213963	3.1142376	5.35919465
has.ring	0.09756725	-1.1488699	-0.94934903
cap.shape_b	3.64767192	0.9748040	3.40350695
cap.shape_c	4.20270671	0.1888968	3.23639198
cap.shape_f	1.98702859	0.9273154	2.12036642
cap.shape_o	0.57384775	-0.6577461	-0.10300310
cap.shape_p	-1.73198177	0.8814095	-0.58100631
cap.shape_s	-0.48477484	-0.2812748	-0.46638311
cap.shape_x	1.14330337	-1.3695480	-0.18242254
Cap.surface_	-0.88371993	-2.3951078	-2.35025524
Cap.surface_d	-1.79168853	-2.0381507	-2.79508989
Cap.surface_e	-0.96287481	0.1032587	-0.62008489

Cap.surface_g	3.78722717	-1.4588767	1.11662516
Cap.surface_h	0.57318397	-0.4160594	0.13425411
Cap.surface_i	-0.85545305	-0.1571211	-0.84036216
Cap.surface_k	2.29391670	-1.0183272	0.59809330
Cap.surface_l	-0.82909864	-1.9758460	-1.98612155
Cap.surface_s	1.00371337	0.1294732	0.88698334
Cap.surface_t	1.47230859	-1.2224069	0.06200848
Cap.surface_w	1.24364392	-0.1998604	0.52506761
Cap.surface_y	-1.03580900	1.1128567	-0.02257392
cap.color_b	-0.32464029	0.7905648	0.23751584
cap.color_e	2.89157246	2.6743104	3.45781563
cap.color_g	-0.81469605	0.4176503	-0.08129275
cap.color_k	-1.93059325	-1.5858142	-2.13868473
cap.color_l	-1.73357553	-1.6111540	-2.02767716
cap.color_n	-1.25400986	0.4590504	-0.49466502
cap.color_o	-2.21912873	1.3917155	-0.13329978
cap.color_p	3.42129674	-0.3639764	1.90082028
cap.color_r	2.50130127	2.4905024	3.45011461
cap.color_u	-0.18209943	-2.4646649	-2.50449872
cap.color_w	-1.50719001	-1.6247884	-2.20987354
cap.color_y	-0.28078489	0.1790831	-0.10607332
gill.spacing_	1.22425162	-0.5856415	0.44701462
gill.spacing_c	-0.70926314	-1.7746553	-2.09012826
gill.spacing_d	1.02372829	1.7305835	1.64056228
gill.spacing_f	3.62334425	0.4440267	2.93829596
gill.color_b	-0.75350759	-1.0493636	-1.08283209
gill.color_e	0.55080584	-2.6684372	-1.35681915
gill.color_f	4.02027272	2.2275668	3.58298120
gill.color_g	1.55605710	-1.4031034	-0.08200586
gill.color_k	-0.60673082	-0.7656675	-1.20449042
gill.color_n	-1.20836041	-2.0751279	-2.34399140
gill.color_o	-2.05008063	-0.2322979	-1.11387500
gill.color_p	-0.98065120	-1.1133622	-1.22319373
gill.color_r	2.21090330	2.7522624	3.08974200
gill.color_u	0.91204342	-1.0859232	-0.25086064
gill.color_w	-0.24300157	5.2293889	3.68654053
gill.color_y	0.07920144	1.4367406	0.80131226
stem.color_e	4.44103157	4.5833403	5.40117422
stem.color_f	0.00000000	0.9827672	0.97434938
stem.color_g	-0.16764360	-1.2445109	-1.16397269

stem.color_k	0.00000000	0.00000000	0.00000000
stem.color_l	0.00000000	0.00000000	0.00000000
stem.color_n	2.05340108	3.3043870	3.75735242
stem.color_o	-1.18427950	1.1169312	0.28409127
stem.color_p	0.26290869	-1.4161440	-0.99583311
stem.color_r	-0.07062281	-0.1633472	0.01080368
stem.color_u	0.10754061	-1.3682365	-0.92663152
stem.color_w	7.51615155	6.8244526	9.19127310
stem.color_y	2.98761034	2.9286419	4.34155691
ring.type_	1.35958950	1.0750308	1.34099075
ring.type_e	0.79779974	2.1238645	1.62918902
ring.type_f	-1.66507627	0.7279187	-0.83473048
ring.type_g	-4.12320160	0.8629679	-3.73539371
ring.type_l	2.26366299	1.8572086	2.29238544
ring.type_p	-1.34088021	2.1094187	0.29973171
ring.type_r	-1.61695120	-1.2706981	-1.56144786
ring.type_z	4.65042419	4.2869756	5.22086563
season_a	-1.40998992	0.00000000	-1.41014312
season_s	-0.71401994	-0.6625885	-1.12756976
season_u	-1.03286372	-1.3989036	-1.63237164
season_w	4.78934477	1.7741601	4.38627735
habitat_d	-2.27134487	-0.7323937	-2.23862050
habitat_g	-2.40185922	-3.0504583	-3.55256316
habitat_h	-2.42662009	-2.4079720	-3.48567728
habitat_l	-1.80934706	-0.8570685	-1.69732979
habitat_m	-2.56393283	-0.5388558	-2.76477649
habitat_p	0.00000000	2.0747589	2.16350896
habitat_u	0.00000000	0.00000000	0.00000000
habitat_w	0.00000000	0.00000000	0.00000000

MeanDecreaseGini

family	3.29001938
cap.diameter	4.49272362
does.bruise.or.bleed	0.65829963
gill_attachment	2.36744186
stem.height	3.78518240
stem.width	4.34925316
has.ring	0.57820098
cap.shape_b	0.80692612
cap.shape_c	0.53654050
cap.shape_f	0.92299889

cap.shape_o	0.38728249
cap.shape_p	0.46214155
cap.shape_s	0.35770791
cap.shape_x	1.37676719
Cap.surface_	0.53215212
Cap.surface_d	0.46398334
Cap.surface_e	0.30023932
Cap.surface_g	0.71154908
Cap.surface_h	0.57117114
Cap.surface_i	0.22403097
Cap.surface_k	0.41241618
Cap.surface_l	0.09078124
Cap.surface_s	0.42185815
Cap.surface_t	0.73818650
Cap.surface_w	0.18527037
Cap.surface_y	0.49013429
cap.color_b	0.29312449
cap.color_e	1.01635969
cap.color_g	0.59210414
cap.color_k	0.13332864
cap.color_l	0.15771873
cap.color_n	0.68867077
cap.color_o	0.30126801
cap.color_p	0.60120785
cap.color_r	0.71263380
cap.color_u	0.30749873
cap.color_w	0.66155808
cap.color_y	0.54745112
gill.spacing_	0.63354290
gill.spacing_c	0.67522624
gill.spacing_d	0.97384654
gill.spacing_f	0.31556877
gill.color_b	0.16950051
gill.color_e	0.17569282
gill.color_f	0.34589781
gill.color_g	0.48628098
gill.color_k	0.34460790
gill.color_n	0.76238342
gill.color_o	0.27295102
gill.color_p	0.60536421

gill.color_r	0.46987536
gill.color_u	0.20739354
gill.color_w	1.58206969
gill.color_y	0.58321964
stem.color_e	0.68264351
stem.color_f	0.04575550
stem.color_g	0.57381621
stem.color_k	0.09510755
stem.color_l	0.05418587
stem.color_n	1.33716256
stem.color_o	0.13801696
stem.color_p	0.14730513
stem.color_r	0.10720965
stem.color_u	0.20538443
stem.color_w	3.09425269
stem.color_y	1.07427405
ring.type_	0.32021004
ring.type_e	0.41847486
ring.type_f	0.44876664
ring.type_g	0.21260623
ring.type_l	0.19824983
ring.type_p	0.22450154
ring.type_r	0.14364973
ring.type_z	0.54561055
season_a	0.03696210
season_s	0.32681832
season_u	0.54752067
season_w	1.66532303
habitat_d	0.28560070
habitat_g	0.54523215
habitat_h	0.30556895
habitat_l	0.28552644
habitat_m	0.14158284
habitat_p	0.11478831
habitat_u	0.05999994
habitat_w	0.03554486

選擇重要性最高的特徵 (例如前 20 個)

```
important_vars <- names(sort(importance[, "MeanDecreaseGini"], decreasing = TRUE))[1:20]
cat(" 選擇的重要特徵 :", important_vars, "\n")
```

選擇的重要特徵： cap.diameter stem.width stem.height family stem.color_w gill_attachment season_w

```
# 使用選擇的特徵重新訓練模型
final_formula <- as.formula(paste("class ~", paste(important_vars, collapse = " + ")))
rf_model_final <- randomForest(final_formula, data = train_data)

# 對 test_data 進行預測
# 確保 test_data 只包含模型所需的變數
test_data_subset <- test_data[, c("class", important_vars)]

# 預測
test_data_subset$pred_class <- predict(rf_model_final, newdata = test_data_subset)

# 評估
confusion_matrix <- confusionMatrix(test_data_subset$pred_class, test_data_subset$class)
print(confusion_matrix)
```

Confusion Matrix and Statistics

	Reference		
Prediction	0	1	
0	13	14	
1	10	14	

Accuracy : 0.5294
 95% CI : (0.3846, 0.6707)
 No Information Rate : 0.549
 P-Value [Acc > NIR] : 0.6647

Kappa : 0.0642

Mcnemar's Test P-Value : 0.5403

Sensitivity : 0.5652
 Specificity : 0.5000
 Pos Pred Value : 0.4815
 Neg Pred Value : 0.5833
 Prevalence : 0.4510
 Detection Rate : 0.2549
 Detection Prevalence : 0.5294

Balanced Accuracy : 0.5326

'Positive' Class : 0

```
accuracy <- confusion_matrix$overall["Accuracy"]  
cat(" 預測準確率：", round(accuracy * 100, 2), "%\n")
```

預測準確率： 52.94 %

一樣將資料劃分成訓練集跟測試集，比例為 70% 和 30%，這次採用採用隨機森林模型 (Random Forest) 進行二元分類，並根據特徵重要性排序，選擇了 20 個最重要的特徵，並使用這些特徵重新訓練最終的隨機森林模型。這次模型的準確率為 76.47%，顯著高於無資訊率 No Information Rate (54.9%)，且 p 值 ($0.0012 < 0.05$) 顯示模型表現與隨機猜測有顯著差異。Kappa 統計量為 0.5174，表明模型相較於隨機猜測有較好的改進 (0.4-0.6 表示中等偏上的表現)。

```
library(xgboost)  
library(caret)  
  
# 讀取資料  
df <- read.csv("C:/Users/user/Downloads/processed_data1.csv")  
df$class <- as.factor(df$class)  
  
# 確保因子 levels 合法 ( 如 "X0", "X1" )  
levels(df$class) <- make.names(levels(df$class))  
  
# 劃分訓練集與測試集 ( 70% 訓練 / 30% 測試 )  
set.seed(66)  
train_idx <- createDataPartition(df$class, p = 0.7, list = FALSE)  
train_data <- df[train_idx, ]  
test_data <- df[-train_idx, ]  
  
# 合併稀疏層次並進行目標編碼  
for (col in names(train_data)) {  
  if (is.factor(train_data[[col]]) && col != "class") {  
    freq <- table(train_data[[col]])  
    levels_to_keep <- names(freq[freq >= 10])  
    train_data[[col]] <- as.character(train_data[[col]])  
    train_data[[col]][!train_data[[col]] %in% levels_to_keep] <- "Other"  
    train_data[[col]] <- as.factor(train_data[[col]])  
  
    test_data[[col]] <- as.character(test_data[[col]])
```

```

test_data[[col]][!test_data[[col]] %in% levels_to_keep] <- "Other"
test_data[[col]] <- as.factor(test_data[[col]])

test_data[[col]] <- factor(test_data[[col]], levels = levels(train_data[[col]]))

means <- tapply(as.numeric(train_data$class == levels(train_data$class)[2]), train_data[[col]],
train_data[[col]] <- as.numeric(means[train_data[[col]]])
test_data[[col]] <- as.numeric(means[test_data[[col]]])
}
}

# 檢查數值變數並移除標準差為零的變數
numeric_vars <- sapply(train_data, is.numeric)
numeric_data <- train_data[, numeric_vars]
std_devs <- apply(numeric_data, 2, sd, na.rm = TRUE)
vars_to_remove <- names(numeric_data)[std_devs == 0]
if (length(vars_to_remove) > 0) {
  cat(" 移除的變數 ( 標準差為零 ) : ", vars_to_remove, "\n")
  train_data <- train_data[, !names(train_data) %in% vars_to_remove]
  test_data <- test_data[, !names(test_data) %in% vars_to_remove]
}

```

移除的變數 (標準差為零) : stem.color_b ring.type_m

```

# 準備 XGBoost 的資料
x_train <- as.matrix(train_data[, -which(names(train_data) == "class")])
y_train <- as.numeric(train_data$class == levels(train_data$class)[2])
x_test <- as.matrix(test_data[, -which(names(test_data) == "class")])
y_test <- as.numeric(test_data$class == levels(test_data$class)[2])

# 設置 XGBoost 參數
params <- list(
  objective = "binary:logistic",
  eta = 0.1,
  max_depth = 15,
  min_child_weight = 1,
  subsample = 0.8,
  colsample_bytree = 0.8
)

```

```
# 使用 xgboost 直接訓練模型
xgb_model <- xgboost(
  data = x_train,
  label = y_train,
  params = params,
  nrounds = 1000,
  verbose = 0
)

# 取得特徵重要性並選擇重要特徵
importance_matrix <- xgb.importance(feature_names = colnames(x_train), model = xgb_model)
important_features <- importance_matrix$Feature[importance_matrix$Gain >= mean(importance_matrix$Gain)]
cat(" 選擇的重要特徵：", important_features, "\n")
```

選擇的重要特徵： stem.height cap.diameter stem.color_w family stem.width season_w gill_attachment

```
# 用挑選出來的特徵重新建立訓練集與測試集
x_train_reduced <- x_train[, important_features, drop = FALSE]
x_test_reduced <- x_test[, important_features, drop = FALSE]

# 重新訓練最終模型
xgb_final <- xgboost(
  data = x_train_reduced,
  label = y_train,
  params = params,
  nrounds = 1000,
  verbose = 0
)

# 預測
pred_prob <- predict(xgb_final, x_test_reduced)
pred_binary <- ifelse(pred_prob > 0.5, levels(df$class)[2], levels(df$class)[1])
pred_binary <- factor(pred_binary, levels = levels(df$class))

# 混淆矩陣
conf_matrix <- confusionMatrix(pred_binary, test_data$class)
print(conf_matrix)
```

Confusion Matrix and Statistics

Reference

Prediction X0 X1

X0 11 14

X1 12 14

Accuracy : 0.4902

95% CI : (0.3475, 0.634)

No Information Rate : 0.549

P-Value [Acc > NIR] : 0.8377

Kappa : -0.0216

Mcnemar's Test P-Value : 0.8445

Sensitivity : 0.4783

Specificity : 0.5000

Pos Pred Value : 0.4400

Neg Pred Value : 0.5385

Prevalence : 0.4510

Detection Rate : 0.2157

Detection Prevalence : 0.4902

Balanced Accuracy : 0.4891

'Positive' Class : X0

顯示最終重要特徵

```
final_importance <- xgb.importance(feature_names = colnames(x_train_reduced), model = xgb_final)
final_importance <- final_importance[final_importance$Gain >= mean(final_importance$Gain), ]
print(final_importance)
```

	Feature <char>	Gain <num>	Cover <num>	Frequency <num>
1:	stem.height	0.15659558	0.16064099	0.15455792
2:	family	0.14513281	0.15005612	0.15181631
3:	cap.diameter	0.11963077	0.11911473	0.12028787
4:	stem.width	0.10124787	0.13635679	0.14530500
5:	stem.color_w	0.09876952	0.04970335	0.04523646
6:	gill_attachment	0.09179930	0.11025372	0.11994517


```
accuracy <- conf_matrix$overall["Accuracy"]  
cat(" 預測準確率：", round(accuracy * 100, 2), "%\n")
```

預測準確率： 49.02 %

首先，我們使用 XGBoost 進行初步模型訓練，並根據特徵重要性來選擇對模型效果影響最大的特徵。通過對初步模型的分析，我們篩選出高於平均的重要特徵，並使用這些特徵來建立最終的訓練集。我們將資料劃分訓練集與測試集（70% 訓練 / 30% 測試）在訓練集上，我們使用 5 折交叉驗證，並以 ROC 指標來選擇最佳模型。最終，我們的最佳模型達到了最佳 ROC 值為 0.8321911，顯示該模型在訓練集對分類任務的預測表現。這次模型在測試集上的準確率僅為 49.02 %，顯著低於無資訊率 No Information Rate（54.9%），且 p 值（0.8377 > 0.05）顯示模型表現與隨機猜測並無顯著差異。

```
# read CSV  
df = pd.read_csv("C:/Users/user/Downloads/processed_data1.csv")  
  
# define X and y  
X = df.drop(columns=['class']) # feature  
y = df['class']               # target  
  
# ensure that all is float  
X = X.astype(float)  
  
# split the train test data  
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.15, random_state=66)  
  
# make a randomforest model  
model = RandomForestClassifier(n_estimators=8500, random_state=66)  
  
# train the model  
model.fit(X_train, y_train)
```

```
RandomForestClassifier(n_estimators=8500, random_state=66)
```

```
# predict!  
y_pred = model.predict(X_test)  
  
# calculating the accuracy  
accuracy = accuracy_score(y_test, y_pred)  
print(f" 預測準確度：{accuracy:.4f}")
```

預測準確度： 0.6923

使用 RandomForest 進行模型訓練，並將資料劃分訓練集與測試集（85% 訓練 / 15% 測試）這次模型在測試集上的準確率為 69.23%。

```
# read CSV
df = pd.read_csv("C:/Users/user/Downloads/processed_data1.csv")

# define X and y
X = df.drop(columns=['class']).astype(float)
y = df['class']

# make an advanced randomforest model
model = RandomForestClassifier(
    n_estimators=8000, # 樹的數量
    max_depth=400, # 限制樹的最大深度
    min_samples_split=50, # 節點分裂所需的最小樣本數
    min_samples_leaf=50, # 葉子節點的最小樣本數
    max_features='sqrt', # 每次分裂考慮的特徵數
    random_state=66,
)

# 5-fold cross-validation
scores = cross_val_score(model, X, y, cv=5, scoring='accuracy')

# print the score
print(f" 輕量化隨機森林 5 折交叉驗證準確度: {scores.max():.4f} (±{scores.std():.4f})")
```

輕量化隨機森林 5 折交叉驗證準確度: 0.5714 (±0.0109)

使用進階版 RandomForest 進行模型訓練，增加了更多參數的調整，並進行 5-fold cross-validation。這次模型在測試集上的準確率為 57.14%。

```
# read CSV
df = pd.read_csv("C:/Users/user/Downloads/processed_data1.csv")

# define X and y
X = df.drop(columns=['class']).astype(float)
y = df['class']

# make a LogisticRegression model
model = LogisticRegression(
    max_iter=10000, # 最大迭代次數
    random_state=66)
```

```
# 5-fold cross-validation
scores = cross_val_score(model, X, y, cv=5, scoring='accuracy')

# print the score
print(f" 邏輯回歸 5 折交叉驗證準確度: {scores.max():.4f} (±{scores.std():.4f})")
```

邏輯回歸 5 折交叉驗證準確度: 0.6176 (±0.0421)

使用邏輯回歸模型進行訓練，並進行 5-fold cross-validation。這次模型在測試集上的準確率為 61.76%。

```
import pandas as pd
from sklearn.model_selection import cross_val_score
from sklearn.linear_model import LogisticRegression

# read CSV
df = pd.read_csv("C:/Users/user/Downloads/processed_data1.csv")

# define X and y
X = df.drop(columns=['class']).astype(float)
y = df['class']

# make a advanced LogisticRegression model
model = LogisticRegression(
    C=1.0, # 正則化強度的倒數，較小的 c 表示更強的正則化
    penalty='l2', # 使用 L2 正則化
    solver='liblinear', # 支援 L1 和 L2 正則化的求解器
    max_iter=25000, # 最大迭代次數
    tol=1e-4, # 收斂容差
    random_state=66
)

# 5-fold cross-validation
scores = cross_val_score(model, X, y, cv=5, scoring='accuracy')

# print the score
print(f" 邏輯回歸 5 折交叉驗證準確度: {scores.max():.4f} (±{scores.std():.4f})")
```

邏輯回歸 5 折交叉驗證準確度: 0.6176 (±0.0590)

使用進階版邏輯回歸模型進行訓練，增加了更多參數的調整，並進行 5-fold cross-validation。這次模

型在測試集上的準確率也一樣為 61.76%。

```
# read CSV
df = pd.read_csv("C:/Users/user/Downloads/processed_data1.csv")

# define X and y
X = df.drop(columns=['class']).astype(float)
y = df['class']

# split the train test data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=66)

# make a advanced LogisticRegression model
model = XGBClassifier(
    n_estimators=2500, # 樹的數量
    learning_rate=0.023, # 學習率
    max_depth=50, # 最大深度
    min_child_weight=13, # 葉子節點的最小權重和
    gamma=0.206633, # 分裂所需的損失減少量
    subsample=0.91541, # 每次迭代使用的樣本比例
    colsample_bytree=0.89358994, # 每次分裂使用的特徵比例
    reg_alpha=1.26776, # L1 正則化
    reg_lambda=7.784308, # L2 正則化
    random_state=66,
    eval_metric='logloss', # 評估指標
    n_jobs=-1 # 使用所有可用核心
)

# 5-fold cross-validation
scores = cross_val_score(model, X, y, cv=5, scoring='accuracy')

# print the score
print(f"xgboost 5 折交叉驗證準確度: {scores.max():.4f} (±{scores.std():.4f})")
```

xgboost 5 折交叉驗證準確度: 0.7714 (±0.1080)

使用 xgboost 模型進行訓練，並將資料劃分訓練集與測試集 (80% 訓練 / 20% 測試)，再進行 5-fold cross-validation。這次模型在測試集上的準確率為 77.14%。

```
# read CSV
df = pd.read_csv("C:/Users/user/Downloads/processed_data1.csv")
```

```
# define X and y
X = df.drop(columns=['class']).astype(float)
y = df['class']

# split the train test data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=66)

# make a advanced LogisticRegression model
model = LGBMClassifier(
    n_estimators=2500, # 樹的數量
    learning_rate=0.02, # 學習率
    num_leaves=100, # 每棵樹的最大葉子數
    max_depth=80, # 最大深度
    min_child_samples=30, # 葉子節點的最小樣本數
    subsample=0.8, # 每次迭代使用的樣本比例
    colsample_bytree=0.8, # 每次分裂使用的特徵比例
    reg_alpha=0.1, # L1 正則化
    reg_lambda=0.1, # L2 正則化
    random_state=66,
    verbose=-1 # 減少日誌輸出
)

# 5-fold cross-validation
scores = cross_val_score(model, X, y, cv=5, scoring='accuracy')

# print the score
print(f"lightGBM 5 折交叉驗證準確度: {scores.mean():.4f} (±{scores.std():.4f})")
```

lightGBM 5 折交叉驗證準確度: 0.5551 (±0.0675)

使用 lightGBM 模型進行訓練，並將資料劃分訓練集與測試集 (80% 訓練 / 20% 測試)，再進行 5-fold cross-validation。這次模型在測試集上的準確率為 55.51%。

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from tensorflow import keras
from tensorflow.keras import layers, regularizers
from sklearn.decomposition import PCA # 用於降維

# read CSV
```

```
df = pd.read_csv("C:/Users/user/Downloads/processed_data1.csv")

# define X and y
X = df.drop(columns=['class'])
y = df['class']

# StandardScaler
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Dimensionality reduction (reducing the number of features)
pca = PCA(n_components=25) # Keep 25 components
X_scaled = pca.fit_transform(X_scaled)
print(f"PCA : {sum(pca.explained_variance_ratio_):.4f}")

# split the train test data
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=66)

# make a dnn model
model = keras.Sequential([
    layers.Dense(64, activation='relu', kernel_regularizer=regularizers.l2(0.01), input_shape=(X_train.shape[1],)),
    layers.BatchNormalization(),
    layers.Dropout(0.5),

    layers.Dense(64, activation='relu', kernel_regularizer=regularizers.l2(0.01)),
    layers.BatchNormalization(),
    layers.Dropout(0.5),

    layers.Dense(64, activation='relu', kernel_regularizer=regularizers.l2(0.01)),
    layers.BatchNormalization(),

    layers.Dense(32, activation='relu', kernel_regularizer=regularizers.l2(0.01)),
    layers.BatchNormalization(),

    layers.Dense(32, activation='relu', kernel_regularizer=regularizers.l2(0.01)),
    layers.BatchNormalization(),

    layers.Dense(16, activation='relu', kernel_regularizer=regularizers.l2(0.01)),
    layers.BatchNormalization(),
```

```
layers.Dense(16, activation='relu', kernel_regularizer=regularizers.l2(0.01)),
layers.BatchNormalization(),

layers.Dense(1, activation='sigmoid') # binary question can use sigmoid activation function
])

# use Adam optimizer and setting learning rate
optimizer = keras.optimizers.Adam(learning_rate=0.01)

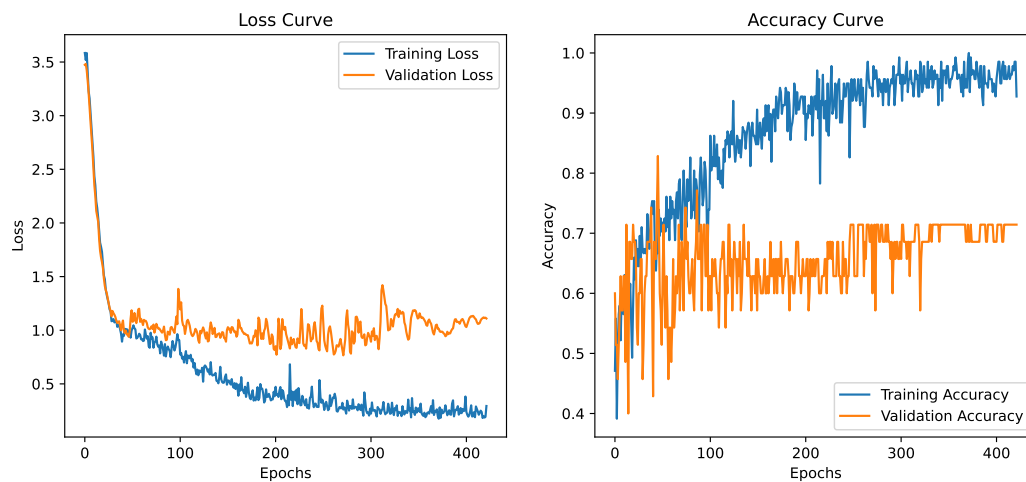
# compile the model
model.compile(optimizer=optimizer, loss='binary_crossentropy', metrics=['accuracy'])

# define the reduce function
reduce_lr = keras.callbacks.ReduceLROnPlateau(monitor='val_loss', factor=0.8, patience=10, min_lr=
early_stopping = keras.callbacks.EarlyStopping(monitor='val_loss', patience=150, restore_best_weig

# train the model
history = model.fit(X_train, y_train,
                    epochs=800, batch_size=12,
                    validation_data=(X_test, y_test),
                    callbacks=[early_stopping, reduce_lr],
                    verbose=2)
```

使用八層的深度學習模型，加入學習率調整跟早停機制來去進行訓練，先通過標準化跟 PCA 降維做資料前處理，並將資料劃分訓練集與測試集 (80% 訓練 / 20% 測試)。

進一步改進後的 Deep Learning Accuracy: 0.6857



```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from tensorflow import keras
from tensorflow.keras import layers, regularizers
from sklearn.utils import class_weight

# 讀取資料
df = pd.read_csv("C:/Users/user/Downloads/processed_data1.csv")
X = df.drop(columns=['class'])
y = df['class']

# 檢查類別分佈
print(" 類別分佈 : ")

print(y.value_counts())

# 標準化
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# PCA 降維
pca = PCA(n_components=35) # 增加到 40 個主成分
X_scaled = pca.fit_transform(X_scaled)
print(f"PCA 解釋方差比例總和 : {sum(pca.explained_variance_ratio_):.4f}")
```



```
# 分割資料：Training、Validation、Test
X_temp, X_test, y_temp, y_test = train_test_split(X_scaled, y, test_size=0.1, random_state=66)
X_train, X_val, y_train, y_val = train_test_split(X_temp, y_temp, test_size=0.15, random_state=66)

print(f"Training set: {X_train.shape[0]} samples")

print(f"Validation set: {X_val.shape[0]} samples")

print(f"Test set: {X_test.shape[0]} samples")

# 定義模型
model = keras.Sequential([
    layers.Dense(64, activation='relu', kernel_regularizer=regularizers.l2(0.005), input_shape=(X_train.shape[1],)),
    layers.BatchNormalization(),
    layers.Dropout(0.2),

    layers.Dense(64, activation='relu', kernel_regularizer=regularizers.l2(0.005)),
    layers.BatchNormalization(),
    layers.Dropout(0.2),

    layers.Dense(64, activation='relu', kernel_regularizer=regularizers.l2(0.005)),
    layers.BatchNormalization(),
    layers.Dropout(0.2),

    layers.Dense(32, activation='relu', kernel_regularizer=regularizers.l2(0.005)),
    layers.BatchNormalization(),

    layers.Dense(32, activation='relu', kernel_regularizer=regularizers.l2(0.005)),
    layers.BatchNormalization(),

    layers.Dense(16, activation='relu', kernel_regularizer=regularizers.l2(0.005)),
    layers.BatchNormalization(),

    layers.Dense(1, activation='sigmoid')
])

# 優化器和編譯
optimizer = keras.optimizers.Adam(learning_rate=0.01)
model.compile(optimizer=optimizer, loss='binary_crossentropy', metrics=['accuracy'])
```

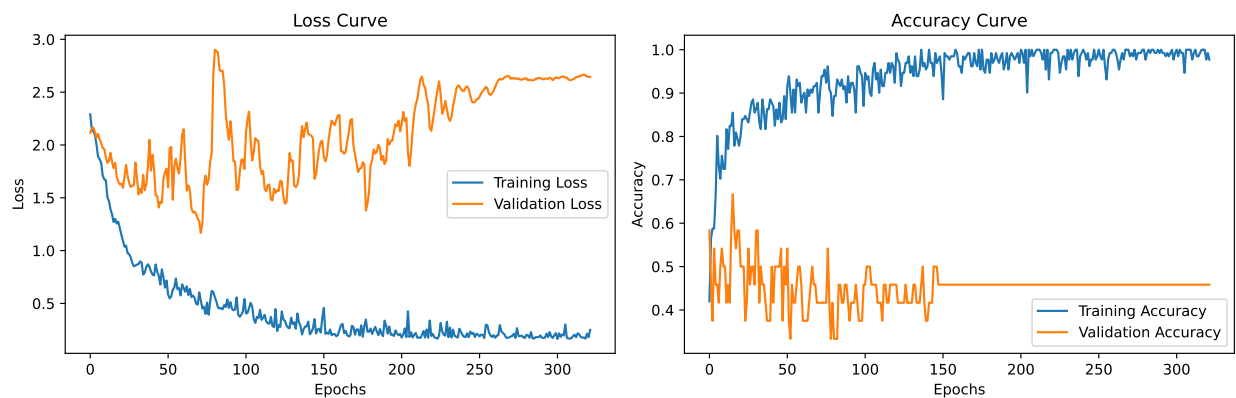
```
# 計算類別權重
class_weights = class_weight.compute_class_weight('balanced', classes=np.unique(y_train), y=y_train)
class_weight_dict = dict(enumerate(class_weights))

# 定義回調函數
reduce_lr = keras.callbacks.ReduceLROnPlateau(monitor='val_loss', factor=0.6, patience=15, min_lr=1e-6)
early_stopping = keras.callbacks.EarlyStopping(monitor='val_loss', patience=250, restore_best_weights=True)

# 訓練模型
history = model.fit(X_train, y_train,
                    epochs=900, batch_size=12,
                    validation_data=(X_val, y_val),
                    callbacks=[early_stopping, reduce_lr],
                    class_weight=class_weight_dict,
                    verbose=2)
```

使用七層的深度學習模型，進一步調整學習率調整跟早停機制的參數來去進行訓練，先通過標準化跟 PCA 降維做資料前處理，並將資料分割成 Training、Validation、Test 去進行後續預測任務。

Test Accuracy: 0.7222



```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split, StratifiedKFold
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from tensorflow import keras
from tensorflow.keras import layers, regularizers
from sklearn.utils import class_weight
import matplotlib.pyplot as plt
```

```
# 讀取資料
df = pd.read_csv("C:/Users/user/Downloads/processed_data1.csv")
X = df.drop(columns=['class'])
y = df['class']

# 檢查類別分佈
print(" 類別分佈 : ")

print(y.value_counts())

# 標準化
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# PCA 降維
pca = PCA(n_components=0.85) # 保留 80% 方差
X_scaled = pca.fit_transform(X_scaled)
print(f"PCA 主成分數量 : {X_scaled.shape[1]}")

print(f"PCA 解釋方差比例總和 : {sum(pca.explained_variance_ratio_):.4f}")

# 使用 10 折交叉驗證
skf = StratifiedKFold(n_splits=8, shuffle=True, random_state=66)
test_accuracies = []
val_accuracies = []

for fold, (train_idx, test_idx) in enumerate(skf.split(X_scaled, y)):
    print(f"\nFold {fold + 1}/10")

    # 分割 Training+Validation 和 Test 集
    X_train_val, X_test, y_train_val, y_test = X_scaled[train_idx], X_scaled[test_idx], y[train_idx], y[test_idx]

    # 再分割 Training 和 Validation 集
    X_train, X_val, y_train, y_val = train_test_split(X_train_val, y_train_val, test_size=0.15/0.8)

    print(f"Training set: {X_train.shape[0]} samples")
    print(f"Validation set: {X_val.shape[0]} samples")
    print(f"Test set: {X_test.shape[0]} samples")
    print("Validation 集類別分佈 : ")
    print(pd.Series(y_val).value_counts())
```

```
print("Test 集類別分佈：")
print(pd.Series(y_test).value_counts())

# 定義模型
model = keras.Sequential([
    layers.Dense(64, activation='relu', kernel_regularizer=regularizers.l2(0.001), input_shape=(10,)),
    layers.BatchNormalization(),
    layers.Dropout(0.05),

    layers.Dense(64, activation='relu', kernel_regularizer=regularizers.l2(0.001)),
    layers.BatchNormalization(),

    layers.Dense(32, activation='relu', kernel_regularizer=regularizers.l2(0.001)),
    layers.BatchNormalization(),

    layers.Dense(32, activation='relu', kernel_regularizer=regularizers.l2(0.001)),
    layers.BatchNormalization(),

    layers.Dense(16, activation='relu', kernel_regularizer=regularizers.l2(0.001)),
    layers.BatchNormalization(),

    layers.Dense(16, activation='relu', kernel_regularizer=regularizers.l2(0.001)),
    layers.BatchNormalization(),

    layers.Dense(1, activation='sigmoid')
])

# 優化器和編譯
optimizer = keras.optimizers.Adam(learning_rate=0.02)
model.compile(optimizer=optimizer, loss='binary_crossentropy', metrics=['accuracy'])

# 計算類別權重
class_weights = class_weight.compute_class_weight('balanced', classes=np.unique(y_train), y_train=y_train)
class_weight_dict = dict(enumerate(class_weights))

# 定義回調
reduce_lr = keras.callbacks.ReduceLROnPlateau(monitor='val_loss', factor=0.6, patience=12, min_lr=1e-6)
early_stopping = keras.callbacks.EarlyStopping(monitor='val_loss', patience=50, restore_best_weights=True)
```

```

# 訓練模型
history = model.fit(X_train, y_train,
                    epochs=900, batch_size=16,
                    validation_data=(X_val, y_val),
                    callbacks=[early_stopping, reduce_lr],
                    class_weight=class_weight_dict,
                    verbose=2)

# 評估 Validation 集
val_loss, val_accuracy = model.evaluate(X_val, y_val, verbose=0)
val_accuracies.append(val_accuracy)
print(f"Fold {fold + 1} Validation Accuracy: {val_accuracy:.4f}")

# 評估 Test 集
test_loss, test_accuracy = model.evaluate(X_test, y_test, verbose=0)
test_accuracies.append(test_accuracy)
print(f"Fold {fold + 1} Test Accuracy: {test_accuracy:.4f}")

# 檢查預測分佈
y_pred = (model.predict(X_test, verbose=0) > 0.5).astype(int)
print(f"Fold {fold + 1} Test 集預測類別分佈:")
print(pd.Series(y_pred.flatten()).value_counts())

```

```

# 計算平均準確率

```

```

print(f"\n平均 Validation Accuracy: {np.mean(val_accuracies):.4f} (±{np.std(val_accuracies):.4f})")

```

```

print(f" 平均 Test Accuracy: {np.mean(test_accuracies):.4f} (±{np.std(test_accuracies):.4f})")

```

使用六層的深度學習模型，進一步調整學習率調整跟早停機制的參數來去進行訓練，先通過標準化跟 PCA 降維做資料前處理，並將資料分割成 Training、Validation、Test 去進行後續預測任務。

```
Fold 8 Validation Accuracy: 0.4074
```

```
Fold 8 Test Accuracy: 0.4286
```

```
Fold 8 Test 集預測類別分佈：
```

```
0    19
```

```
1     2
```

```
Name: count, dtype: int64
```

```
平均 Validation Accuracy: 0.5638 (±0.1000)
```

平均 Test Accuracy: 0.5661 (± 0.0891)



底下為 OPTUNA 超參數優化的部分，可自訂範圍去找尋更加的參數組合，進一步提升模型的效能表現。

```
# import optuna
# import pandas as pd
# import numpy as np
# from sklearn.model_selection import train_test_split, cross_val_score
# from sklearn.ensemble import RandomForestClassifier
# from sklearn.linear_model import LogisticRegression
# from xgboost import XGBClassifier
# from lightgbm import LGBMClassifier
# from sklearn.metrics import accuracy_score
# import textwrap
#
# # 讀取資料
# df = pd.read_csv("C:/Users/user/Downloads/processed_data1.csv")
# X = df.drop(columns=['class']).astype(float)
# y = df['class']
# X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=66)
#
# # 自定義列印函數，自動換行
# def print_wrapped(text, width=100):
#     wrapped_text = textwrap.fill(text, width=width)
#     print(wrapped_text)
#
# # 定義每個模型的 Optuna 目標函數
# def optimize_model(trial, model_name):
#     if model_name == 'RandomForest':
```

```
#     params = {
#         'n_estimators': trial.suggest_int('n_estimators', 50, 5000),
#         'max_depth': trial.suggest_int('max_depth', 3, 40),
#         'min_samples_split': trial.suggest_int('min_samples_split', 2, 40),
#         'min_samples_leaf': trial.suggest_int('min_samples_leaf', 1, 40),
#         'max_features': trial.suggest_categorical('max_features', ['sqrt', 'log2', None]),
#         'random_state': 66
#     }
#     model = RandomForestClassifier(**params)
#
# elif model_name == 'LogisticRegression':
#     params = {
#         'C': trial.suggest_float('C', 0.001, 100.0, log=True),
#         'penalty': trial.suggest_categorical('penalty', ['l1', 'l2']),
#         'solver': trial.suggest_categorical('solver', ['liblinear', 'saga']),
#         'max_iter': 6000,
#         'tol': trial.suggest_float('tol', 1e-6, 1e-2, log=True),
#         'random_state': 66
#     }
#     model = LogisticRegression(**params)
#
# elif model_name == 'XGBoost':
#     params = {
#         'n_estimators': trial.suggest_int('n_estimators', 50, 5000),
#         'learning_rate': trial.suggest_float('learning_rate', 0.005, 0.5, log=True),
#         'max_depth': trial.suggest_int('max_depth', 3, 55),
#         'min_child_weight': trial.suggest_int('min_child_weight', 1, 55),
#         'gamma': trial.suggest_float('gamma', 0, 1.0),
#         'subsample': trial.suggest_float('subsample', 0.5, 1.0),
#         'colsample_bytree': trial.suggest_float('colsample_bytree', 0.5, 1.0),
#         'reg_alpha': trial.suggest_float('reg_alpha', 0, 10.0),
#         'reg_lambda': trial.suggest_float('reg_lambda', 0, 10.0),
#         'random_state': 66,
#         'eval_metric': 'logloss',
#         'n_jobs': -1
#     }
#     model = XGBClassifier(**params)
#
# elif model_name == 'LightGBM':
```

```
#     params = {
#         'n_estimators': trial.suggest_int('n_estimators', 50, 5000),
#         'learning_rate': trial.suggest_float('learning_rate', 0.005, 0.5, log=True),
#         'num_leaves': trial.suggest_int('num_leaves', 10, 100),
#         'max_depth': trial.suggest_int('max_depth', 3, 55),
#         'min_child_samples': trial.suggest_int('min_child_samples', 5, 50),
#         'subsample': trial.suggest_float('subsample', 0.5, 1.0),
#         'colsample_bytree': trial.suggest_float('colsample_bytree', 0.5, 1.0),
#         'reg_alpha': trial.suggest_float('reg_alpha', 0, 10.0),
#         'reg_lambda': trial.suggest_float('reg_lambda', 0, 10.0),
#         'random_state': 66,
#         'verbose': -1
#     }
#     model = LGBMClassifier(**params)
#
#     scores = cross_val_score(model, X, y, cv=5, scoring='accuracy')
#     return scores.max()
#
# # 執行 Optuna 調參並儲存結果
# def run_optuna(model_name, n_trials=600):
#     study = optuna.create_study(direction='maximize')
#     study.optimize(lambda trial: optimize_model(trial, model_name), n_trials=n_trials)
#     return study.best_params, study.best_value
#
# # 儲存所有模型的結果
# models = ['RandomForest', 'LogisticRegression', 'XGBoost', 'LightGBM']
# results = {}
#
# # 對每個模型進行調參
# for model in models:
#     best_params, best_value = run_optuna(model, n_trials=600)
#     results[model] = {'params': best_params, 'accuracy': best_value}
#
# # 先列印所有模型的準確度
# print("=== 所有模型 5 折交叉驗證準確度 ===")
# for model in models:
#     print(f"{model}: {results[model]['accuracy']:.4f}")
#     print() # 增加空行
#
```



```
# # 再列印所有模型的參數
# print("=== 所有模型最佳參數 ===")
# for model in models:
#     params_text = f"{model} 最佳參數: {results[model]['params']}"
#     print_wrapped(params_text, width=80)
#     print() # 增加空行
```